

1 Solution — GIAM 6.2

Problem 2

1.1 Problem

Consider the relation A defined by

$$A = \{(x, y) : x \text{ has the same astrological sign as } y\}.$$

Is A symmetric or anti-symmetric? Explain.

1.2 Setup

Let P be a population of people, and let

$$\sigma : P \rightarrow Z$$

assign to each person their astrological sign, where Z is the set of twelve zodiac signs (Aries, Taurus, ..., Pisces). Then

$$A = \{(x, y) \in P \times P : \sigma(x) = \sigma(y)\}.$$

In other words, A is the relation “shares a sign,” *pulled back from equality on Z* by the function σ .

The two properties at issue, for a relation R on a set X :

- **Symmetric.** For all $x, y \in X$, $(x, y) \in R \implies (y, x) \in R$.
- **Antisymmetric.** For all $x, y \in X$, $[(x, y) \in R \text{ and } (y, x) \in R] \implies x = y$.

1.3 A Is Symmetric

If $(x, y) \in A$, then $\sigma(x) = \sigma(y)$. Equality on Z is symmetric, so $\sigma(y) = \sigma(x)$, hence $(y, x) \in A$. The implication holds for every pair, so A is **symmetric**.

(A plain-language version: “ x has the same sign as y ” and “ y has the same sign as x ” express the same fact.)

1.4 A Is Not Antisymmetric

Pick any two distinct people who share a sign — say two Pisces, Alice and Bob. Then $\sigma(\text{Alice}) = \sigma(\text{Bob})$, so

$$(\text{Alice}, \text{Bob}) \in A \quad \text{and} \quad (\text{Bob}, \text{Alice}) \in A,$$

yet $\text{Alice} \neq \text{Bob}$. The implication " $(x, y) \in A$ and $(y, x) \in A \implies x = y$ " fails, so A is **not antisymmetric**.

The same argument works under any sign with at least two members in P , which is the rule rather than the exception for any realistic population.

1.5 Remark — A Is an Equivalence Relation

Same-sign-as inherits all three equivalence properties from equality on Z :

- **Reflexive.** $\sigma(x) = \sigma(x)$, so $(x, x) \in A$.
- **Symmetric.** Shown above.
- **Transitive.** If $\sigma(x) = \sigma(y)$ and $\sigma(y) = \sigma(z)$, then $\sigma(x) = \sigma(z)$.

The induced partition of P has at most twelve blocks — one per zodiac sign — given by the fibers $\sigma^{-1}(s)$ for $s \in Z$. This is the standard recipe: every function $f : X \rightarrow Y$ defines an equivalence relation on X via $x \sim x' \iff f(x) = f(x')$, with equivalence classes equal to the fibers of f .

1.6 Remark — Symmetric and Antisymmetric Together

A relation that is *both* symmetric and antisymmetric must be a subset of the diagonal: for any $(x, y) \in R$, symmetry gives $(y, x) \in R$, and antisymmetry then forces $x = y$. So

$$\text{symmetric and antisymmetric} \iff R \subseteq \Delta_X.$$

For A to be antisymmetric, then, every person would have to be the only one with their sign — impossible once $|P| > 12$. Symmetry and antisymmetry are nearly opposites: a relation can satisfy *both* only when it is trivial (a sub-relation of equality), so seeing rich symmetry as in A is already a strong signal that antisymmetry will fail.

This is the structural complement of the previous problem: “smarter than” is *asymmetric* (and so vacuously antisymmetric, not symmetric); “same astrological sign as” is symmetric and definitively not antisymmetric.